Limitation of Command Line Argument :

As we know by using Command Line Argument, we can pass some value at runtime. These values are stored in String array variable and then only the execution of the program will be started.
The basic limitation of Command line Argument is, we can't ask to enter the value from our end user as shown in the Program.

Program :

```
package com.rani.command_line_argument;
public class ReadCharacter
{
    public static void main(String[] args)
    {
        System.out.println("Enter your gender [M/F]");

        char ch = args[0].charAt(0);
        System.out.println("Your gender is :"+ch);
    }
}
```

Note : In the above program, after providing the gender value, It is asking for Gender which is not a recommended way.

Note : charAt(int indexPosition) is a predefined non static method of String class, which is used to retrieve a character
from the given String based on the index position.

How to take the input from the user :

* There are so many ways to take the input from the user which are as follows :

a) java.io.DataInputStream
b) java.io.BufferedReader
c) java.lang.System.in.read()
d) java.io.Console (used to read password)
e) java.util.Scanner

How to read the client data using Scanner class :

* Scanner is a predefined class available in java.util package from JDK 1.5V.
* It is used to read the client data in **more appropriate way by using user friendly message.**

static field of System class :

* System class has provided the following static fields :

1) System.out : It is used to print normal message on the console window.

2) System.err : It is used to print error message on the console window. [Red color]

3) System.in : It is used to take the input from the source.

How to create an object for Scanner class :

* Scanner class provides various non static methods, In order to call and execute these methods, we need to create the object for Scanner class.

Scanner sc = new Scanner(System.in);

Non static methods of Scanner class :

1) public String next() : Used to read a single word.

2) public String nextLine() : Used to read multiple words or complete line.

3) public byte nextByte() : Used to read byte value.

4) public short nextShort() : Used to read short value.

5) public int nextInt() : Used to read int value.

6) public long nextLong() : Used to read long value.

7) public float nextFloat() : Used to read float value.

8) public double nextDouble() : Used to read double value.

9) public boolean nextBoolean() : Used to read boolean value.

10) public char next().charAt(0) : Will read a single character.

WAP to read your name from end user :

```
import java.util.*;

public class ReadName
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your name :");
        String name = sc.nextLine();
        System.out.println("Your name is :"+name);
    }
}
```

Note : java.lang is a default package so we need not to import but java.util package is **not a default package so, to use the classes which are available in java.util package we must import the package.**

```
package com.rani.scanner_demo;

import java.util.Scanner;

public class ReadCharacter
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your gender [M/F] :");
        char gender = sc.next().charAt(0);
        System.out.println("Your gender is :"+gender);
        sc.close();

    }
}
```

```
package com.rani.scanner_demo;

import java.util.Scanner;

public class ReadScannerData
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your roll :");
        int roll = Integer.parseInt(sc.nextLine());

        System.out.print("Enter your Name :");
        String name = sc.nextLine();

        System.out.print("Enter your Fees :");
        double fees = Double.parseDouble(sc.nextLine());

        System.out.println("Your Roll number is :"+roll);
        System.out.println("Your Name is :"+name);
        System.out.println("Your Fees is :"+fees);

        sc.close();
    }
}
```

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4) Punctuator :

It is also called separators.

It is used to inform the compiler how things are grouped in the code.

( ) { } [ ] ; , . @

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5) Operators

It is a symbol which describes that how a calculator will be performed on operands.

Types Of Operators :

Operator Precedence

Operator	Precedence
postfix	expr++ expr--
unary	++expr --expr +expr -expr ~ !
multiplicative	* / %
additive	+ -
shift	<< >> >>>
relational	< > <= >= instanceof
equality	== !=
bitwise AND	&
bitwise exclusive OR	^
bitwise inclusive OR	|
logical AND	&&
logical OR	||
ternary	? :
assignment	= += -= *= /= %= &= ^= |= <<= >>= >>>=